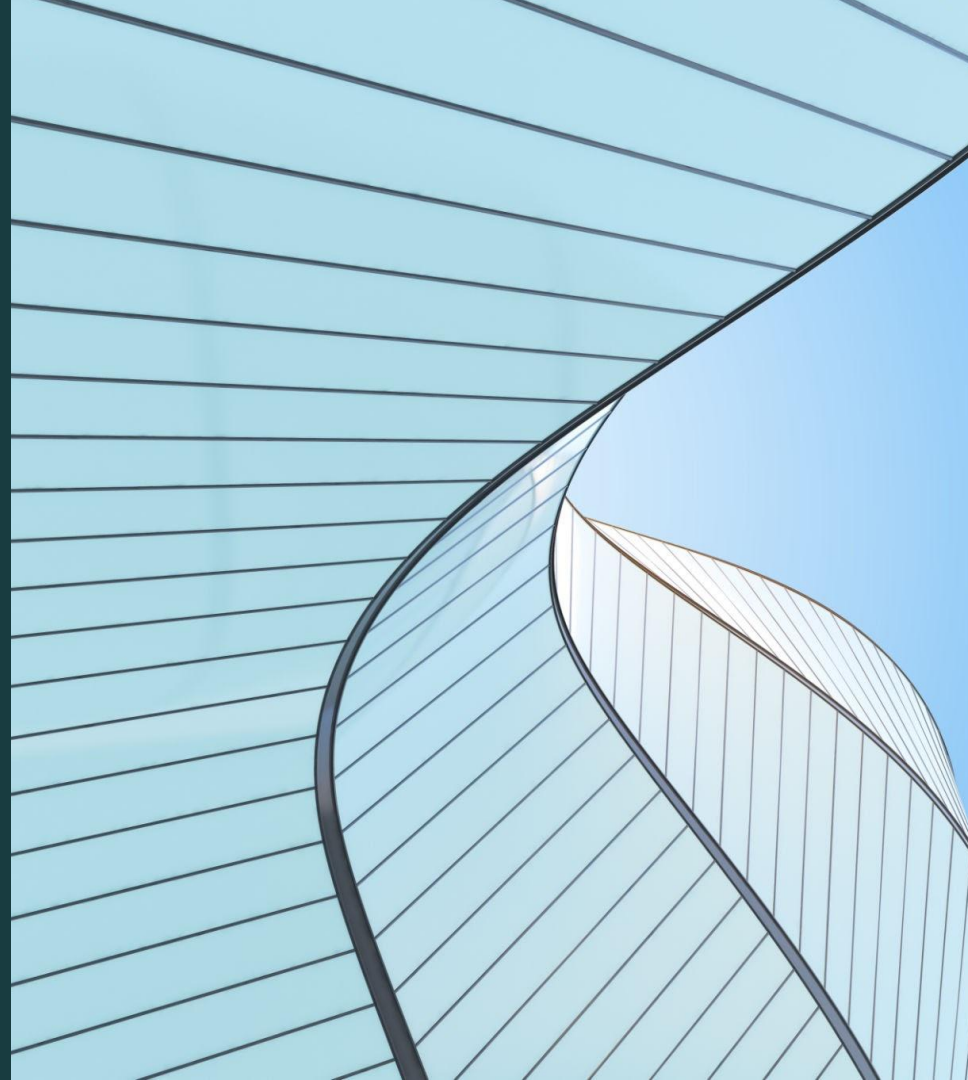


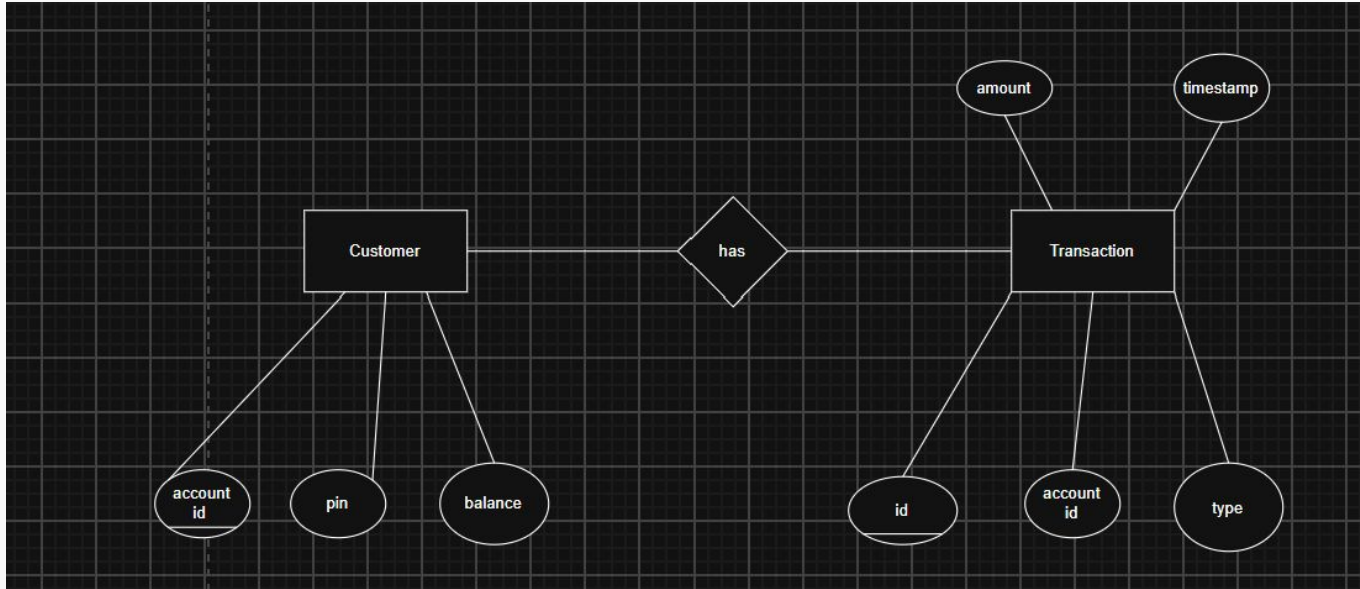
Muhammad Amer

CLI BANK

Bank of CLI is a Java-based banking application that allows customers to register, log in, check their balance, deposit and withdraw money, transfer funds, and view transaction history. It uses a PostgreSQL database to store account information and transactions.



ERD



Programming Language

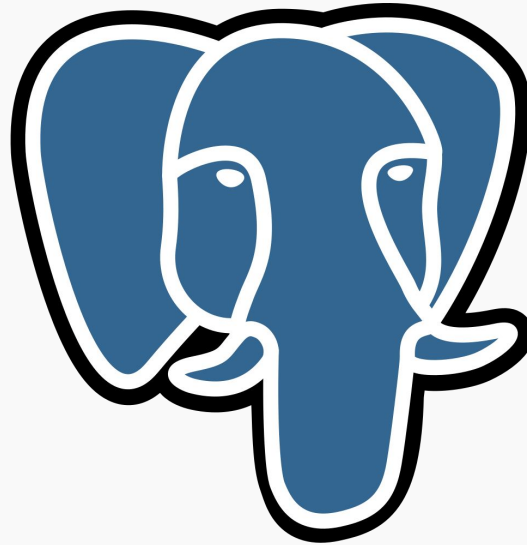
JAVA



The main programming language used to build the application. We used Java to implement the banking operations, business logic, and terminal interface.

Database

PostgreSQL



A relational database used to store customer accounts, balances, and transaction records. It allows the information to remain saved after the application closes.

Connecting to database

JDBC



Java Database Connectivity allows Java to communicate with PostgreSQL. We used JDBC in our DAO classes to execute SQL queries and update database records.

BUILD TOOL

MAVEN



A build and dependency management tool for Java projects. We used it to manage libraries, compile the application, and run our JUnit tests.

TESTING

JUnit 5



A Java testing framework used to verify that individual methods behave as expected. We used it to test successful and unsuccessful login attempts.

Object-Oriented Programming

Encapsulation

```
public class Customer {  
  
    private int accountId;  
    private int pin;  
    private double balance;  
  
    public int getAccountId() {  
        return accountId;  
    }  
  
    public double getBalance() {  
        return balance;  
    }  
  
    public void setBalance(double balance) {  
        this.balance = balance;  
    }  
}
```

Abstraction

```
public interface CustomerService {  
    boolean login(int accountId, int pin);  
    double getBalance(int accountId);  
    boolean deposit(int accountId, double amount);  
    boolean withdraw(int accountId, double amount); }
```

Inheritance

```
public class CustomerServiceImpl
    implements CustomerService {

    @Override
    public boolean login(int
accountId, int pin) {
        // Login logic
    }
}
```